Programming Worlds

PROGRAMMING WORLDS ACADEMY

Git & GitHub 101

From Zero to Your First Pull Request

BEYOND CODE. BEYOND LIMITS.

02

MODULE 02

Installing and Configuring Git

Install Git and VS Code, create your GitHub account, and tell Git who you are.

MODULE 02 · LESSON 2.1

Installing Git and VS Code

تثبيت Git وVS Code

Learning objectives

- Install Git on Windows.
- Install Visual Studio Code.
- Open the integrated terminal and choose Git Bash.
- Verify the installation with `git --version`.

Prerequisites

Module 1. A Windows computer and an internet connection.

Introduction

Before any command works, the tools must be on your computer. This lesson installs both, then checks that Git answers.

Explanation

Install Git. Go to `git-scm.com` and choose **Download for Windows**. Run the installer. The default options are safe for beginners. You may choose Visual Studio Code as the default editor, and choose **main** as the name for new branches if the installer offers it. Click Next until Install, then Finish.

Verify. Open **Git Bash** from the Start menu and run `git --version`. If Git answers with a version number, it is installed.

Install VS Code. Go to `code.visualstudio.com`, download the Windows installer and run it. Keep **Add to PATH** enabled.

Open the terminal in VS Code. Use the menu **Terminal → New Terminal**, or press `Ctrl+`` . In the terminal panel, use the dropdown arrow next to the **+** and choose **Git Bash** as the profile.

Real-world analogy

> Installing tools is like stocking a kitchen before you cook: nothing to cook yet, but every recipe in this course needs these tools.

Diagram

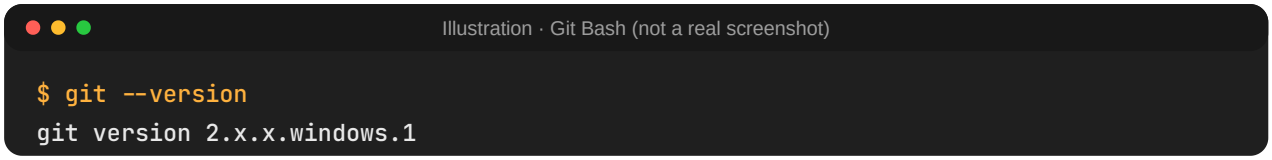


Figure 1 · Illustration of the expected output

Commands

Run this in Git Bash or the VS Code terminal. `--version` is an option that asks the program to print its version and exit.

```
git --version
```

Common mistakes and solutions

“git is not recognized”	Close and reopen the terminal. If it still fails, run the Git installer again.
Opening PowerShell and expecting Bash commands.	Choose Git Bash in the terminal profile list.
Skipping Add to PATH for VS Code.	Re-run the installer and enable it.

Practical exercise

> Install Git and VS Code, open a terminal inside VS Code, select Git Bash, and run `git --version`.

Expected result

The terminal prints a line starting with `git version`.

Knowledge check

1. Which command checks the Git installation?

- ○ git check
- ○ git –version
- ○ git start

2. Which terminal do we use in this course?

- ○ Git Bash
- ○ Notepad
- ○ Paint

3. Which shortcut opens a terminal in VS Code?

- ○ Alt+F4
- ○ Ctrl+`
- ○ Ctrl+Z

Answers

- git –version — `git --version` prints the installed version.
- Git Bash — Git Bash gives us the same commands on Windows.
- Ctrl+` — Ctrl+` toggles the integrated terminal.

Summary

You installed Git and VS Code, opened a terminal in VS Code, chose Git Bash, and confirmed Git works.

Key vocabulary

Terminal	A text window for typing commands
Git Bash	A Bash terminal that comes with Git for Windows
PATH	The list of folders Windows searches for programs

Next lesson

Lesson 2.2 — Creating your GitHub account. The tools are ready. Next you create your online home.

MODULE 02 · LESSON 2.2

Creating your GitHub account

إنشاء حساب GitHub

Learning objectives

- Create and verify a GitHub account.
- Find your way around the dashboard.
- Secure the account with two-factor authentication.
- Name the ways Git signs in to GitHub.

Prerequisites

Lesson 2.1 and an email address you can open.

Introduction

Your GitHub account is your identity online. Your projects, your history and your future portfolio live there, so we protect it from day one.

Explanation

Create the account. Go to `github.com` and choose **Sign up**. Enter your email, a strong password and a username. Choose a username you would be happy to show an employer.

Verify your email. GitHub sends a code to your email. Enter it to finish. You may also be asked to solve a short puzzle.

The dashboard. After sign-in you see your repositories, activity and suggestions. Your avatar (top right) opens the menu with **Your profile**, **Your repositories** and **Settings**.

Profile. Add a photo and a short bio. Keep personal details you do not want public out of it.

Security. Use a unique password in a password manager. In **Settings → Password and authentication** enable **two-factor authentication (2FA)** with an authenticator app, and save the recovery codes somewhere safe.

How Git signs in. Over **HTTPS**, Git asks for your GitHub login through a sign-in window and keeps it in Git Credential Manager (included with Git for Windows). With **SSH**, you add a key pair to your account. Never use your GitHub password as a Git password in the terminal. You will set up a method in Module 5.

Real-world analogy

> Your account is like a bank account for your code: a strong password and a second lock (2FA) keep it yours.

Diagram

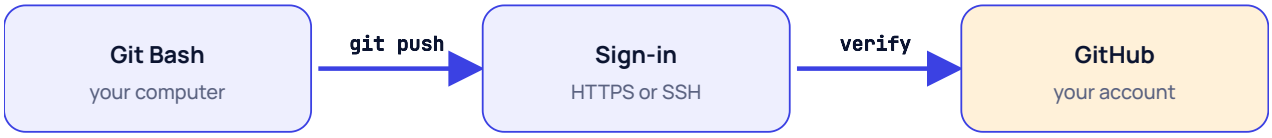


Figure 1 · Git proves who you are before it uploads

Commands

No Git commands here. All steps happen in the browser.

Common mistakes and solutions

Losing the 2FA recovery codes.	Store them in a password manager or print them.
Using a private email in public commits.	Later you can use the GitHub “noreply” address (Lesson 2.3).
Choosing a joke username.	Pick a professional one. It becomes part of your URLs.

Practical exercise

> Create your account, verify the email, add a profile photo, and turn on 2FA.

Expected result

You can open your profile at `github.com/your-username` and Settings shows 2FA as enabled.

Knowledge check

1. Why enable 2FA?

- ○ It is required to install Git
- ○ A second check protects the account
- ○ It makes Git faster

2. Where do your projects live online?

- ○ GitHub
- ○ VS Code
- ○ Git Bash

3. Name one way Git signs in to GitHub.

- ○ Screenshots
- ○ SSH keys
- ○ Email attachments

Answers

- A second check protects the account — 2FA stops access with only a stolen password.
- GitHub — GitHub hosts your remote repositories.
- SSH keys — HTTPS with Credential Manager and SSH keys are the two common methods.

Summary

You created and secured a GitHub account and learned the two sign-in methods Git uses: HTTPS and SSH.

Key vocabulary

2FA	A second proof of identity at sign-in
HTTPS / SSH	Two ways Git connects to GitHub
Dashboard	Your home page on GitHub

Next lesson

Lesson 2.3 — First-time Git configuration. Now you tell Git who you are.

MODULE 02 · LESSON 2.3

First-time Git configuration

إعداد Git لأول مرة

Learning objectives

- Set your name, email and default branch.
- Check your settings.
- Explain global and local configuration.
- Separate commit identity from GitHub sign-in.

Prerequisites

Lessons 2.1 and 2.2.

Introduction

Every commit records who made it. Git cannot guess this, so you set it once.

Explanation

Set your identity. Run the first two commands below with your own details. Use the email of your GitHub account, or the private "noreply" address that GitHub shows in **Settings → Emails**.

Default branch. The third command makes new repositories start with a branch named `main`.

Check. `git config --global --list` prints every global setting.

Global vs local. `--global` saves in your user file and applies to all repositories. Without it, inside a repository, a setting is **local** and applies only there. Local wins over global.

Identity vs sign-in. Name and email are only labels stamped on commits. They do not prove who you are. **Authentication** (HTTPS login or SSH key) is what lets you push. You can use SSH later: `ssh-keygen -t ed25519 -C "you@example.com"` creates a key pair. We finish that in Module 5.

Real-world analogy

> The name and email are the signature on a letter. Anyone can write a signature, but only your key opens the mailbox.

Diagram

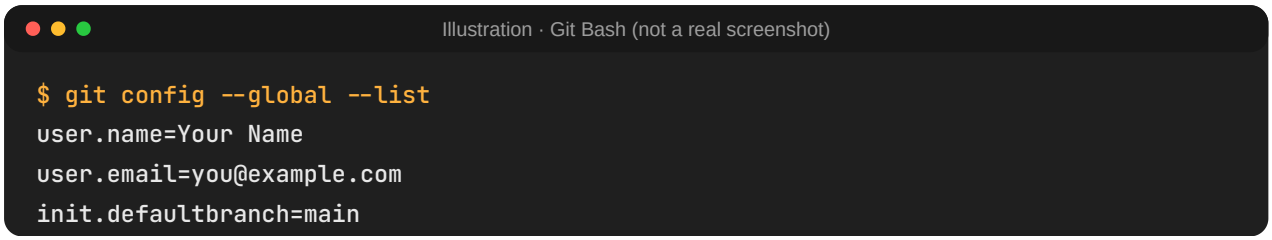


Figure 1 · Illustration of the expected output

Commands

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global init.defaultBranch main
git config --global --list
```

Options: `--global` user-wide setting; `--list` show settings.

Common mistakes and solutions

Typing the quotes wrongly.	Use straight quotes around names with spaces.
Using a different email than GitHub.	Commits may not link to your profile. Use your GitHub email.
Confusing config with login.	Config only labels commits. Sign-in is separate.

Practical exercise

> Run the four commands with your details, then check the output matches what you typed.

Expected result

The list shows your `user.name`, `user.email` and `init.defaultbranch=main`.

Knowledge check

1. What does user.name do?

- ○ Renames the project
- ○ Labels your commits with your name
- ○ Logs you in to GitHub

2. Which wins, local or global?

- ○ Local
- ○ Global
- ○ Neither

3. Does user.email authenticate you?

- ○ Only on Windows
- ○ No, authentication is separate
- ○ Yes

Answers

- Labels your commits with your name — It is stamped on each commit.
- Local — A local setting overrides a global one in that repository.
- No, authentication is separate — It is a label. HTTPS login or SSH key authenticates.

Summary

You set your name, email and default branch, verified them, and learned that commit identity is not the same as GitHub sign-in.

Key vocabulary

Global / Local	All repos / one repo
Authentication	Proving who you are to GitHub
SSH key	A key pair used instead of a password

Next lesson

Module 3 — Creating your first repository. With Git configured, you will create your first project.

MODULE 02

Practical lab 2 · Configure and verify Git

1. Open Git Bash and run `git --version`.
2. Run the three `git config --global` commands with your details.
3. Run `git config --global --list`.
4. Check each line against what you typed.

Check	Pass if…
Version	Output starts with `git version`
Identity	Name and email match your GitHub account
Branch	`init.defaultbranch=main` appears

Evaluation: 3 of 3 checks = pass.

Module 2 quiz (pass mark: 70%)

1. Which command prints your global settings?

- ○ git show config
- ○ git list
- ○ git config –global –list

2. Which tool is the editor with built-in Git?

- ○ Git Bash
- ○ VS Code
- ○ GitHub

3. What should you do with 2FA recovery codes?

- ○ Store them safely
- ○ Share them
- ○ Delete them

4. What does –global mean?

- ○ Uploads to GitHub
- ○ Applies to all your repositories
- ○ Applies to one file

5. Does user.name prove who you are?

- ○ Yes
- ○ Only with 2FA
- ○ No, it is only a label

Answers

- git config –global –list — `--list` prints the settings.
- VS Code — VS Code includes Git buttons.
- Store them safely — They restore access if you lose your device.
- Applies to all your repositories — It saves in your user-level Git file.
- No, it is only a label — Authentication proves identity; name is a label.